

Introduction to Probability and Statistics

Solutions to the exercises from lecture 1

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Exercise 1: Use the data in the moodle materials folder to plot the absolute, relative, and normalized histograms of (choose one)

- a) The latency for communication in our Starlink setup (file: Starlink_latency.csv)
- b) The temperature measured by a sensor during a PhD course at Oulu University (file: temperature.csv)

Are these datasets approximately normal?

Solution: The normalized histograms are shown in Fig. 2, along with with the probability density function (PDF) of the Gaussian distributions with appropriate means and variances calculated from the sample (i.e., sample mean and variance). These plots show that neither the latency nor temperature measurements follow a Gaussian distribution.

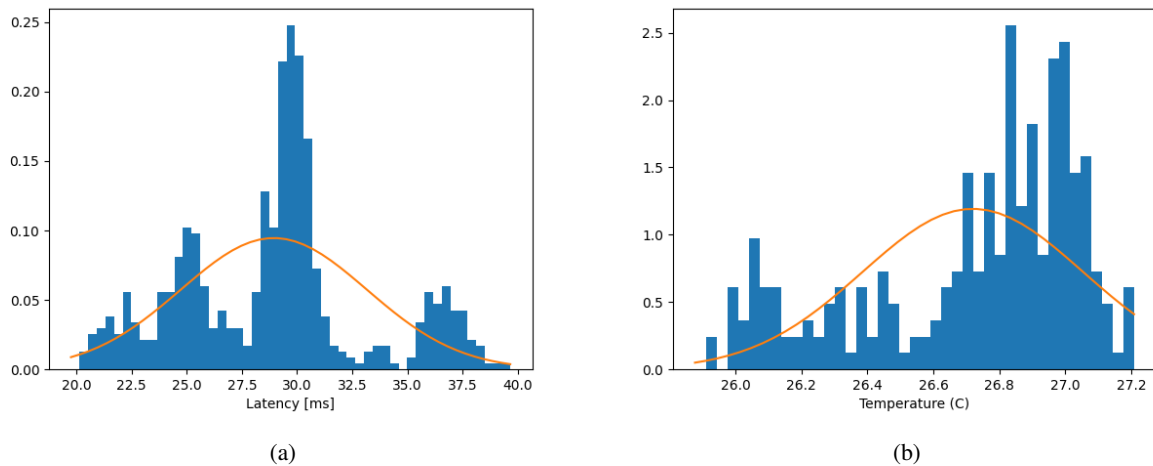


Fig. 1. Normalized histograms for (a) the latency with Starlink and (b) the temperature at the PhD course, along with the PDF of the Gaussian distributions with appropriate means and variances.

Exercise 2: Simulate rolling a fair dice n times or roll it yourself.

- Calculate the sample mean $\bar{X}_i$ and variance S_i^2 for all $i = 1, 2, \dots, n$.
- Plot the sample mean $\bar{X}_i$ and $\bar{X}_i \pm S_i$.
- Plot a normalized histogram with the n outcomes. How many times do you need to roll the dice so the histogram resembles the pmf of a uniform RV?

Tip: You can use the code `dice_loln.py` in the moodle page as a base

Solution: The sample mean $\bar{X}_i$ and variance S_i^2 for $i = \{1, 2, \dots, 1000\}$ rolls of a 6-sided dice ($a = 1$ and $b = 6$) and the normalized histogram of the outcomes. As it can be seen, the sample mean $\bar{X}_i$ tends to the theoretical mean of the distribution $\mu = (a + b)/2 = 3.5$ as i increases and so does S_i to $((b - a + 1)^2 - 1)/12 = 35/12$. Consequently, $\bar{X}_i \pm S_i$ tends to 3.5 ± 1.707 . Nevertheless, even with 1000 experiments, the normalized histogram is not flat.

Exercise 3: Consider the case of flipping a fair coin n times and let H_n be the random variable (RV) of the number of *heads*. Also recall that the sum of n Bernoulli RVs is a Binomial random variable with mean np and variance $np(1-p)$. Calculate and compare the probability of observing $H_{10} = 7$ and $H_{100} = 70$ using each of the following methods.

- Using the formula for the probability mass function (pmf) of Binomial RVs

$$P(H_n = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad (1)$$

- Using the central limit theorem

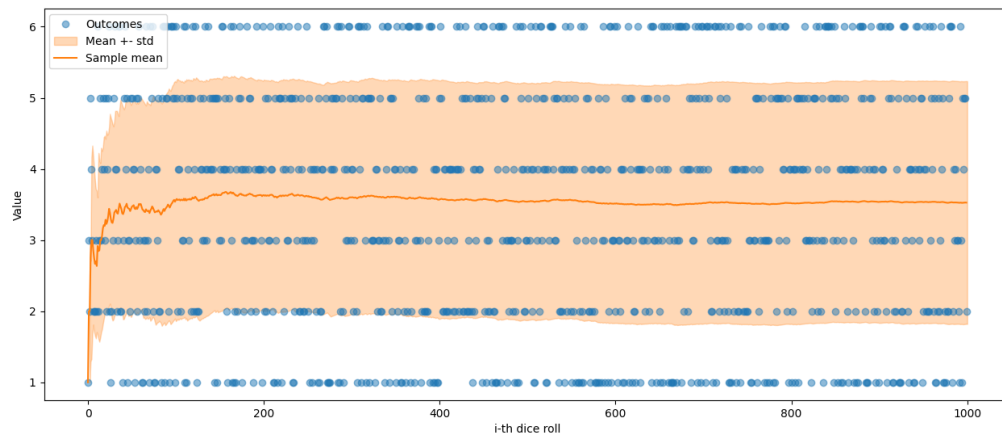
Now calculate and compare the probability of observing $H_{10} \geq 7$ and $H_{100} \geq 70$ with the following methods.

- Summing over the formula for the pmf of Binomial RVs

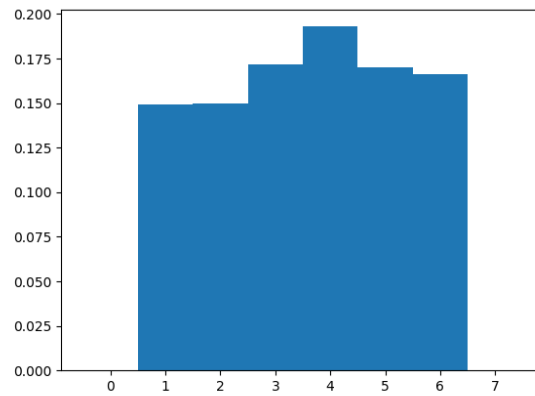
$$P(H_n \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i} \quad (2)$$

- Using the central limit theorem

Solution: a) By setting $n = 10$ and $k = 7$ we get $P(H_{10} = 7) = \binom{10}{7} p^7 (1-p)^{10-7} = 0.117$. Similarly, for $n = 100$ and $k = 70$ we get $P(H_{100} = 70) = 2.3 \times 10^{-5}$. So, observing 70% of the tosses being heads is likely with 10 coin tosses but extremely rare with 100 coin tosses.



(a)



(b)

Fig. 2. (a) Sample mean and variances for $i = \{1, 2, \dots, 1000\}$ and (b) normalized histogram for $n = 1000$.

b) In continuous RVs, the probability of observing a specific outcome is 0.

c) By using the formula for the Binomial distribution is $P(H_{10} \geq 7) = 0.1718$ and $P(H_{100} \geq 70) = 0.00003925$. This shows that having 70% or more of the tosses being heads is likely with 10 coin tosses but is extremely rare with 100 coin tosses.

d) By using the normal approximation, we get $P(H_{10} \geq 7) = 0.1029$ and $P(H_{100} \geq 70) = 0.00003167$.

By comparing the results with these two methods, the absolute error is greater for H_{10} than for H_{100} . This suggests that the normal approximation becomes more accurate as n increases.

Exercise 4: A football team will play 60 games this year. Thirty-two of these games are against

teams playing the champions league, denoted as class A teams, and 28 are against other teams, denoted as class B teams. The outcomes of the games are independent. The team will win each game against a class A team with probability 0.5, and it will win each game against a class B team with probability 0.7. Let X denote its total number of victories in the year.

- Is X a Binomial RV?
- Let X_A and X_B denote, respectively, the number of victories against class A and class B teams. What are their distributions?
- What is the relationship between X , X_A and X_B ?
- Approximate the probability that the team wins 40 or more games this year

Solution:

- No, it is not, since we have two success probabilities $p_A = 0.5$ and $p_B = 0.7$.
- The distributions are $X_A \sim \text{Binom}(n_A, p_A)$ and $X_B \sim \text{Binom}(n_B, p_B)$.
- The relationship is $X = X_A + X_B$
- The variance of a binomial distributions with parameters n and p is $np(1 - p)$. Then, we assume that the distribution of X_A and X_B can be approximated by normal distributions $N_A \sim N(n_A p_A, n_A p_A(1 - p_A))$ and $N_B \sim N(n_B p_B, n_B p_B(1 - p_B))$. Therefore, the sum of N_A and N_B can be approximated $X \sim N(\mu_X, \sigma_X^2)$, where

$$\mu_X = n_A p_A + n_B p_B$$

$$\sigma_X^2 = n_A p_A(1 - p_A) + n_B p_B(1 - p_B).$$

With these, we approximate

$$P(X \geq 40) = 1 - \frac{1}{2} \left[1 + \text{erf} \left(\frac{40 - \mu_X}{\sigma_X \sqrt{2}} \right) \right] = 0.1187$$

Therefore, the probability that this team wins 40 or more gains is quite low.